

Model of Big Bang Nucleosynthesis

Anvel (Marguerite) de Sainte Marie d'Agneaux

1 Introduction

The Big Bang created a large part of the lighter elements which still exist today in the Universe. In the first few milliseconds to seconds, it created a soup of extremely light particles, such as neutrinos, positrons or electrons. These light particles paired and unpaired to form neutrons and protons during a few milliseconds, while the temperature was rapidly dropping and the Universe expanding. In these first instant, protons and neutrons were rapidly exchanging particles, becoming alternatively one or the other. Once the temperature had fallen enough for them to stabilise, the neutrons and protons also started reacting together. These slightly heavier particles paired up into even larger particles, deuterium, tritium, helium, lithium and berillium. While most of the lithium and berillium decayed into helium, traces amount were still made in these times. This was the Big Bang nucleosynthesis. The heavier elements had to wait for stars to start their own nucleosynthesis to exist (Copi et al., [1995](#)).

It is however not possible to observe this event directly. The only way we have, and might ever have, to understand the workings of this nucleosynthesis is by modeling it. Knowing the possible reactions between elements and their reaction rates today, especially at extremely high temperatures, we can make models of the way the Big Bang happened and what it created (Copi et al., [1995](#)).

2 Methods

The reactions between elements which occurred during the Big Bang can be modelled as a set of differential equations, in which positive terms are constructive, and negative ones are destructive. Each equation describes the evolution in time of one element, as a function of the rates of the different reactions, and the density of the elements involved in these reactions. This is a system of differential equations, which can be solved using linear algebra. We will describe the algebraic process in part [Derivation](#), and the reactions, reaction rates and the other variables of the system in [Big Bang Nucleosynthesis](#).

2.1 Derivation

The terms in this set of equation are

$$\frac{dn_j}{dt},$$

n_j being the number density of an element. We will call each of these terms $f_j(\vec{n})$, where j is the type of particle this equation concerns. Then,

$$f'_j(\vec{n}_i)(n_{i+1} - \vec{n}_i) = f_j(\vec{n}_i),$$

where i is a function of time, and describes different steps in the equation. If we change this to concern a set of equations instead of a singular one, we obtain :

$$A'(\vec{n}_i)(n_{i+1} - \vec{n}_i) = A(\vec{n}_i).$$

A is a matrix, which contains the set of linear equations f_j at i , and $A'_j(\vec{n}_i)$ depends on the Jacobian of the system, which is a matrix in which each term is

$$\frac{d}{dn_k} \left(\frac{\partial n_j}{\partial t} \right),$$

j and k can be equal. $A'_j(\vec{n}_i)$ is such that :

$$A'(\vec{n}_i) = |I - dt \cdot J(n_i)|.$$

I is the identity matrix, of the same size as J , and dt is the time step taken between each points n_i . This is a system of linear equations, which can be solved after each times step dt , as:

$$\begin{aligned} A'(\vec{n}_i)(n_{i+1} - \vec{n}_i) &= A(\vec{n}_i), \\ \Leftrightarrow n_{i+1} &= \vec{n}_i + \frac{A(\vec{n}_i)}{A'(\vec{n}_i)}. \end{aligned}$$

For a better visualisation, A is the matrix :

$$A = \begin{bmatrix} \frac{dn_0}{dt} \\ \cdot \\ \cdot \\ \cdot \\ \frac{dn_N}{dt} \end{bmatrix},$$

in which each function defines a different element in the system, and J , if we do not write the dt contribution, is the matrix:

$$\begin{bmatrix} \frac{dn_0}{dn_0} & \cdots & \frac{dn_0}{dn_N} \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot \\ \frac{dn_N}{dn_0} & \cdots & \frac{dn_N}{dn_N} \end{bmatrix}$$

2.2 Big Bang Nucleosynthesis

Each equation in this system is of the shape :

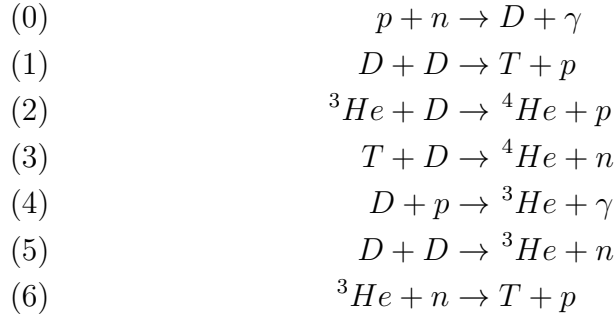
$$\frac{dn_j}{dt} = \sum_l \sum_k n_l n_k \lambda_{lk} - n_j \sum_k n_k,$$

where positive terms are creative terms, which yield more of the element, and negative ones are destructive, changing the element into something else. The λ s are the reaction rates, and indicate how much of each element is created/destroyed per time steps. We also transform these equations into mass fractions instead of densities, to avoid large numbers and make calculations easier. We now have :

$$\frac{dY_j}{dt} = \frac{1}{\rho N_A} (\sum_l \sum_k n_l n_k \lambda_{lk} - n_j \sum_k n_k),$$

where ρ is the total mass density in the system, and N_A is Avogadro's number $N_A = 6.02 \cdot 10^{23} \text{ mol}^{-1}$. Because of this, the total density must be in $\text{mol}/\text{cm}^{-3}$, for the equations to yield the right units.

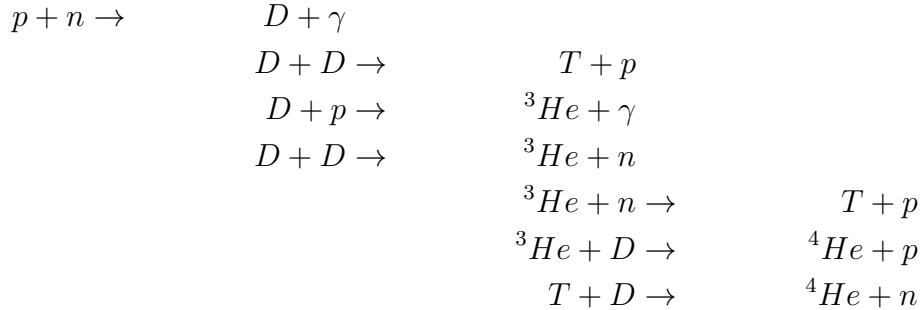
The reactions used in this model are such:



They were chosen as the most mentioned in literature referencing Big Bang nucleosynthesis, starting from reactions between protons and neutrons, and not including elements heavier than helium (Baring, 2024; Navas et al., 2024; Ryan, 2008). We also include neutron decay, as :



Those reactions are sorted by most referenced, but the actual processus of creation and destruction would be better described in this way:



To help in the programming, each particle was given an index (see table 1). Similarly, the reaction rates were given indexes, which are already indicated beside the reactions (2.2). Together, they formed a system of equations which describes this system. For example, the time derivative of the quantity of protons (particle 0) depends

Particle	Index
p	0
n	1
D	2
γ	3
T	4
${}^3\text{He}$	5
${}^4\text{He}$	6

Table 1: Indexing of the elements represented in this system.

on five of the seven reactions, two destructive (0 and 4), and three constructive (1, 2 and 6) :

$$\begin{aligned}
\frac{dY_0}{dt} &= \frac{1}{\rho N_A} (-\lambda_0 n_0 n_1 + \lambda_1 n_2^2 + \lambda_2 n_5 n_2 + \lambda_6 n_5 n_1 - \lambda_4 n_2 n_0 + \lambda_9 n_2) \\
\frac{dY_1}{dt} &= \frac{1}{\rho N_A} (-\lambda_0 n_0 n_1 + \lambda_3 n_4 n_2 + \lambda_5 n_2^2 - \lambda_6 n_5 n_1 - \lambda_9 n_2) \\
\frac{dY_2}{dt} &= \frac{1}{\rho N_A} (\lambda_0 n_0 n_1 - \lambda_1 n_2^2 - \lambda_2 n_5 n_2 - \lambda_3 n_4 n_2 - \lambda_4 n_2 n_0 - \lambda_5 n_2^2) \\
\frac{dY_3}{dt} &= \frac{1}{\rho N_A} (\lambda_0 n_0 n_1 + \lambda_4 n_2 n_0) \\
\frac{dY_4}{dt} &= \frac{1}{\rho N_A} (\lambda_1 n_2^2 - \lambda_3 n_4 n_2 + \lambda_6 n_5 n_1) \\
\frac{dY_5}{dt} &= \frac{1}{\rho N_A} (-\lambda_2 n_5 n_2 + \lambda_5 n_2^2 + \lambda_4 n_2 n_0 - \lambda_6 n_5 n_1) \\
\frac{dY_6}{dt} &= \frac{1}{\rho N_A} (\lambda_2 n_5 n_2 + \lambda_3 n_4 n_2)
\end{aligned} \tag{1}$$

As the Jacobian of the system is derived from this, and it is quite a long derivation, it can be found in [Appendix A](#). Finally, the reaction rates and initial densities are the only elements missing from this system. The reaction rates are taken from Wagoner et al., 1967, except for the neutron decay into protons, which is taken as the inverse of its half-life, which is about 888 seconds (Navas et al., 2024).

Those rates depend on the temperature of the system and on the total density. At this time, temperature decreased with time, as shown in eq. 2. It also varied with the inverse of the expansion rate of the Universe $T \propto R(t)^{-1}$ (Kolb, 1990).

$$T(t) = \frac{1.5 \cdot 10^{10}}{t^{1/2}} \tag{2}$$

The time ranged from $t = 0.01\text{s}$ to $t=100\text{s}$, as that corresponds to the temperature of the Universe when protons and neutrons paired to form deuterium and helium (Kolb, 1990).

The last elements missing to run this model are the initial densities. As they are not yet formed at $t=0.1$, we can already know that the number densities of anything larger than the neutrons and protons will be 0. The only exception could be the γ

Reaction	Rate
(0)	$2.5 \cdot 10^4$
(1)	$3.9 \cdot 10^8 \cdot T_9^{-2/3} \cdot \exp(-4.26 \cdot T_9^{-1/3}) \cdot (1 + 0.0979 \cdot T_9^{1/3} + 0.642 \cdot T_9^{2/3} + 0.440 \cdot T_9)$
(2)	$2.6 \cdot 10^9 \cdot T_9^{-3/2} \cdot \exp(-2.99 \cdot T_9^{-1})$
(3)	$1.38 \cdot 10^9 \cdot T_9^{-3/2} \cdot \exp(-0.745 \cdot T_9^{-1})$
(4)	$2.23 \cdot 10^3 \cdot T_9^{-2/3} \cdot \exp(-3.72 \cdot T_9^{-1/3}) \cdot (1 + 0.112 \cdot T_9^{1/3} + 3.38 \cdot T_9^{2/3} + 2.65 \cdot T_9)$
(5)	$3.9 \cdot 10^8 \cdot T_9^{-2/3} \cdot \exp(-4.26 \cdot T_9^{-1/3}) \cdot (1 + 0.0979 \cdot T_9^{1/3} + 0.642 \cdot T_9^{2/3} + 0.440 \cdot T_9)$
(6)	$7.06 \cdot 10^8$
(7)	$1/800$

Table 2: Reaction rates in the system

particles, but as they are not destroyed, only created, we can still set it to 0 without affecting our system. Remains the proton and neutron density. We set them to be $n_0 = 0.85 \cdot$ and $n_1 = 0.15 \cdot$, as the ratio of proton to neutron was about seven to one (Schramm & Turner, 1998), and the total density was about $2 \cdot 10^{-31} g/cm^3$ (Copi et al., 1995), which, as the molar mass of neutron and proton is about one, gives us $\rho = 2 \cdot 10^{-31}$.

Finally, density also changed with time. As described above, we know the dependence of the expansion rate of the Universe on time, since we know the relation between temperature and time, and temperature and the expansion rate. We also know that $\rho \propto 1/V$ and $V \propto R^3$. Using the equation we have for the temperature, we can set :

$$R(t) = \frac{\sqrt{t}}{4 \cdot 10^{10}}$$

$$\rho(t) = \frac{3\rho_0}{4\pi R(t)}$$

3 Results

3.1 Simulations

3.2 Solving the system differently

In this model, we solved the equation system by deriving the Jacobian by hand, analytically, then solving the system using matrix inversion. It is also possible to find the Jacobian at each time step numerically. Both methods have their uses in different situations.

The analytical method is exact, there is no errors in the values calculated at any points, but it can not be used for every system of equations. Sometimes the functions are not derivable, and it is then not possible to solve them analytically. It is also very demanding ahead of the computation, and the errors can slip in not as an uncertainty in the values but as missing factors, wrong numbers and such. This system was already composed of 7×7 derivatives. This number scales with the number of elements involved, and the derivatives themselves get longer, and thus the risk of an analytical error increases with each new reaction. A system such as the one used in Wagoner

et al., 1967, which has more than eighty reactions and their reverse, is so large that solving it analytically seems impossible.

On the other hands, numerical solutions are not exact. They take a small step Δx in a direction and find an approximation of the analytical function for the value after that step. If it is too much of an approximation, or if the step is too large, the uncertainty also grows. This can lead to extremely large errors in the final results. However, as the entire set of functions only needs to be inputted in a program which does the calculations automatically, if the program is well made and controls the step and approximation to reduce the error, it could theoretically compute an infinite amount of reactions. In the case of the modelisation of astrophysical phenomena, numerical solution also allow for more realistic conditions. Idealised conditions, such as thermodynamic equilibrium, can be solved analytically, but as soon as the system is no longer ideal, numerical solutions might be necessary. Another problem with numerical solution is that they are much more demanding in terms of computation time and power. As it requires an iteration to find a solution at each step, a large set of equations and large amount of steps can take very long. In some cases, long enough that deriving the analytical solution and plugging this into the machine is a better solution.

In the case of this system, with a relatively small amount of equations, all of them derivable, but also quite simple to solve numerically, both solutions work well. The analytical requires a bit more work ahead of the computation to find the derivatives, but the computation itself is easier and not very demanding. The numerical requires less work ahead, but for the amount of steps we took (sometimes more than a million), it might take much longer than the analytical to compute.

Here, we are mixing both, by solving the Jacobian analytically, and then solving the system of differential equations numerically at each time steps, using the implicit Euler method and solving this new system using matrix inversion.

3.3 Adaptative step size

To resolve the system to the best resolution, a constant step size could be a problem. If it is too large, it risks missing important features or creating large errors. On the contrary, if it is too small, it risks looping eternally on slowly evolving features. An adaptative step size which depends on the rate of change of the system seems to be the best option. To solve the system, we already need to calculate $I - dt \cdot J$. A way of adapting the step size would be to aim for this to be as small as possible. If it bellow 0, then the step size is too big. If it is above, then it is too small. Thus a good way of adapting the step size would be to decide on a tolerance and an initial value for dt , and then to update this initial size as a function of the sign of $I - dt \cdot J$. A way to update it would be to halve it, or to take two third of the previous value at each iteration. It require slightly more computer power, as it adds to the iterations, but with a language such as fortran, it remains quite reasonable.

4 Discussion and Conclusion

While this model represents the bare bones of the first part of the Big Bang nucleosynthesis, the difference shown between other models, this model, and observed densities

can be explained by the simplifications made here. In particular, a number of reactions were not included. Wagoner et al., 1967 uses more than eighty reactions and their reverse, for example. We also do not model any reverse reaction, which influences the densities we find.

The code used and a L^AT_EX version of this report can be found at , in "Project"

References

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5 Appendices

5.1 Appendix A

The Jacobian is composed of 7×7 functions, each term being the derivative of $\frac{dn_i}{dt}$ as a function of n_j , where i can be equal to j.

The terms in the first row are :

$$\begin{aligned} J_{00} &= -\lambda_0 n_1 - \lambda_4 n_2 \\ J_{01} &= -\lambda_0 n_0 + \lambda_6 n_5 \\ J_{02} &= 2\lambda_1 n_2 + \lambda_2 n_5 - \lambda_4 n_0 \\ J_{03} &= 0 \\ J_{04} &= 0 \\ J_{05} &= \lambda_2 n_2 + \lambda_6 n_1 \\ J_{06} &= 0 \end{aligned}$$

Second row:

$$\begin{aligned}J_{10} &= -\lambda_0 n_1 \\J_{11} &= -\lambda_0 n_0 - \lambda_6 n_5 \\J_{12} &= \lambda_3 n_4 + 2\lambda_5 n_2 \\J_{13} &= 0 \\J_{14} &= \lambda_3 n_2 \\J_{15} &= -\lambda_6 n_1 \\J_{16} &= 0\end{aligned}$$

Third row:

$$\begin{aligned}J_{20} &= \lambda_0 n_1 - \lambda_4 n_2 \\J_{21} &= \lambda_0 n_0 \\J_{22} &= -2\lambda_1 n_2 - \lambda_2 n_5 - \lambda_3 n_4 - \lambda_4 n_0 - 2\lambda_5 n_2 \\J_{23} &= 0 \\J_{24} &= -\lambda_3 n_2 \\J_{25} &= -\lambda_2 n_2 \\J_{26} &= 0\end{aligned}$$

Fourth row:

$$\begin{aligned}J_{30} &= \lambda_0 n_1 + \lambda_4 n_2 \\J_{31} &= \lambda_0 n_0 \\J_{32} &= \lambda_4 n_0 \\J_{33} &= 0 \\J_{34} &= 0 \\J_{35} &= 0 \\J_{36} &= 0\end{aligned}$$

Fifth row:

$$\begin{aligned}J_{40} &= 0 \\J_{41} &= \lambda_6 n_5 \\J_{42} &= 2\lambda_1 n_2 - \lambda_3 n_4 \\J_{43} &= 0 \\J_{44} &= -\lambda_3 n_2 \\J_{45} &= \lambda_6 n_1 \\J_{46} &= 0\end{aligned}$$

Sixth row:

$$J_{50} = \lambda_4 n_2$$

$$J_{51} = -\lambda_6 n_5$$

$$J_{52} = -\lambda_2 n_5 + 2\lambda_5 n_2 + \lambda_4 n_0$$

$$J_{53} = 0$$

$$J_{54} = 0$$

$$J_{55} = -\lambda_2 n_2 - \lambda_6 n_1$$

$$J_{56} = 0$$

Seventh row:

$$J_{60} = 0$$

$$J_{61} = 0$$

$$J_{62} = \lambda_2 n_5 + \lambda_3 n_4$$

$$J_{63} = 0$$

$$J_{64} = \lambda_3 n_2$$

$$J_{65} = \lambda_2 n_2$$

$$J_{66} = 0$$